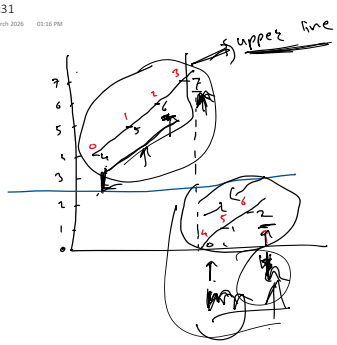
Lec31

25 March 2026 01:16 PM



target = 0

Lo = 0

hi = 6

8 > 4 && 4 < mid

$\frac{0+6}{2} = 3$

```
while(Lo <= hi){
    mid = (Lo+hi)/2
    if(arr[mid] == target){
        return mid
    }
    else if(arr[mid] >= arr[Lo]){
        if(target >= arr[Lo] && target < arr[mid])
            hi = mid-1
        }
        else{
            Lo = mid+1
        }
    }
    else if(target > arr[mid] && target <= arr[hi]){
        L = mid+1
    }
    else{
        hi = mid-1
    }
}
return -1
```

mid = 0

Lo = 4

A B

A

2 = 5/2

candidate = 0

count = 0 1 0 1

3 > 2

for (vote : arr){

if (count == 0){

candidate = vote

}

if(vote == candidate){

count++

}

else{

count--

[2, 2, 1, 1, 1, 2, 2]

A A B

B

5 - 3

2 + 2

8/2

4

6

$\boxed{h = 2}$

else {

count --

}

}

4

[2, 1, 3, 1, 2, 3, 1, 2]

0 1 2 3 4 5 6 7

8

2 = n/3

2 = 3 ✓

1 = 3 ✓

3 = 2

6 2

3 = 9/3